FINM3013 Introduction to Financial Derivatives

Chapter 6 Exercises

1. It is July 30, 2015. The cheapest-to-deliver bond in a September 2015 Treasury bond futures contract is a 13% coupon bond, and delivery is expected to be made on September 30, 2015. Coupon payments on the bond are made on February 4 and August 4 each year. The term structure is flat, and the rate of interest with semiannual compounding is 12% per annum. The conversion factor for the bond is 1.5. The current quoted bond price is $110. Under the actual/actual day count convention, calculate the quoted futures price for the contract.

2. It is June 25, 2014. The futures price for the June 2014 bond futures contract is 118-23. Assume the face value of a bond is $100 and the coupons are paid semiannually.
   a) Calculate the conversion factor for a bond maturing on January 1, 2030, paying a coupon of 10%.
   b) Calculate the conversion factor for a bond maturing on October 1, 2035, paying coupon of 7%.
   c) Suppose that the quoted prices of the bonds in (a) and (b) are 169.00 and 136.00, respectively. Which bond is cheaper to deliver?
   d) Assuming that the cheapest to deliver bond is actually delivered on June 25, 2014, what is the cash price received for the bond? (Assume the coupon payments are on January 1 and July 1.)

3. A portfolio manager plans to use a Treasury bond futures contract to hedge a bond portfolio over the next three months. The portfolio is worth $100 million and will have a duration of 4.0 years in three months. The futures price is 122, and each futures contract is on $100,000 of bonds. The bond that is expected to be cheapest to deliver will have a duration of 9.0 years at the maturity of the futures contract. What position in futures contracts is required?
   a) What adjustments to the hedge are necessary if after one month the bond that is expected to be cheapest to deliver changes to one with a duration of seven years?
   b) Suppose that all rates increase over the three months, but long-term rates increase less than short-term and medium-term rates. What is the effect of this on the performance of the hedge?